# Keio University: Building Canvas LMS on Google Cloud in just three months to provide a cutting-edge e-learning environment

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## About Keio University

Keio University is a private university with origins in a school for Dutch studies. Founded by Fukuzawa Yukichi in 1858, it is Japan's oldest private comprehensive school, and serves students from elementary school up to university and graduate school. With over 160 years of history, Keio University now boasts six campuses, including Mita Campus in Tokyo and Hiyoshi Campus in Kanagawa Prefecture, as well as several satellite campuses. In total, approximately 34,000 students study and 6,000 faculty members are engaged in education,

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research, medicine, and related work at Keio*.

*as of April 1, 2022

Industries: Education

Location: Japan

SQL for PostgreSQL

Bownet

Bow Netsystems Corporation

## Google Cloud results

- Scaled automatically to support a learning environment that accommodates increased access due to the rapid growth of online classes
- Implemented Canvas LMS in just three months with no downtime thanks to

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## About Bow Netsystems

Established in 2000, Bow Netsystems specializes in designing and developing cloud-based applications for education. The company provides the "Bownet Cloud LXAP" SaaS (Software as a Service), an IT platform for education consisting of Canvas LMS and other cutting-edge learning systems.

Google Cloud (https://cloud.google.com/)

Google Cloud Armor (https://cloud.google.com/armor/)

Bow Netsystems

Cloud Filestore (https://cloud.google.com/filestore/)

Cloud SQL for PostgreSQL (https://cloud.google.com/sql/docs/postgres/)

Netsystems

Container Registry (https://cloud.google.com/container-registry/)

**Google Cloud's dedicated support team**

- **Seamlessly links with business systems, and other systems that have already adopted Google Cloud**

Cloud Load Balancing (https://cloud.google.com/load-balancing/)

Compute Engine (https://cloud.google.com/compute/)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Cloud Memorystore (https://cloud.google.com/memorystore/)

"Keio University (https://www.keio.ac.jp/en/) had been operating its own LMS on-premises for some time, but in the knowledge that it would become increasingly important in the future, we decided to move to a world-standard LMS with higher performance and modern features. It was just as we had begun these considerations that COVID-19 appeared, and we suddenly found ourselves in a situation where we had to implement it right away."

Imahori Ryuzaburo, Chief of the Information Technology Center (ITC), Keio University, recalls the above with a wry smile. At the time, many universities in Japan were operating their LMS on-premises, but these were unable to cope with changes in the environment. Not only did the rapid shift to online classes place a great burden on their existing system, but it highlighted other issues, such as missing necessary features. In fact, providing support to faculty staff and students for their proprietary LMS by themselves was also a major pain point.

"An LMS migration typically takes a year or more, but we were no longer in a situation where we could support that. We consulted Bow Netsystems, whom we had already been in discussion with about Canvas LMS, and started the project with the aim of introducing the system in September 2020, just three months later, when classes for the fall semester began. It was then that we also decided to move our infrastructure from on-premises to the cloud," explained Imahori. "Although we hadn't had particular issues with outages on our proprietary LMS, it would have been near-impossible to maintain it going forward, so we thought that a cloud situation would be better able to handle seasonal fluctuations in load."

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*—Imahori Ryuzaburo, Chief, Information Technology Center (ITC), Keio University*

Canvas LMS was originally optimized for a different cloud platform, and is typically built on that platform. Takaharu Takeuchi, ITC Head office Chief of Secretariat, gave the following reasons for choosing Google Cloud.

"In fact, apart from the Canvas LMS implementation, we had plans to move our business systems from on-premises to the cloud. In that communication, we had the impression that Google was the most understanding of the education industry and would be there for us. Also, a number of our faculty members have had some involvement with Google and Google Cloud, and that helped drive our adoption of Google Cloud."

Involvement here includes the international joint research project "Application of AI for Social Good" conducted by Keio University, the Association of Pacific Rim Universities (APRU), and Google from 2019 to 2020; the "COVID-19 Infection Forecasting (Japanese version)"[1] in which Keio University supervised the model design and verification of forecast data for the development of the Japanese version of the model; and the reliability verification of a large-scale survey to model the spread of COVID-19 infection[2]. In addition, Keio University has established a cooperative relationship with Google in many other respects, including the adoption of Google Apps for Education as its information sharing platform in 2014.

[1] See "COVID-19 感染予測 (日本版) の公開について" [COVID-19 Forecast (Japan Edition)] Google Cloud Japan Blog, November 2020 [in Japanese]

[2] See "Google supports COVID-19 AI and data analytics projects," Google Blog, September 2020

*From the left of the photo:*
*・Keio University Information Technology Center (ITC) Office Manager Takaharu Takeuchi*
*・Ryuzaburo Imahori, Chief of Information Technology Center (ITC), Keio University*
*・Haruhiko Mitsuya, Representative Director of Bow Netsystems Corporation*
*・Yuki Ishikawa, System Engineer, Bow Netsystems Corporation*

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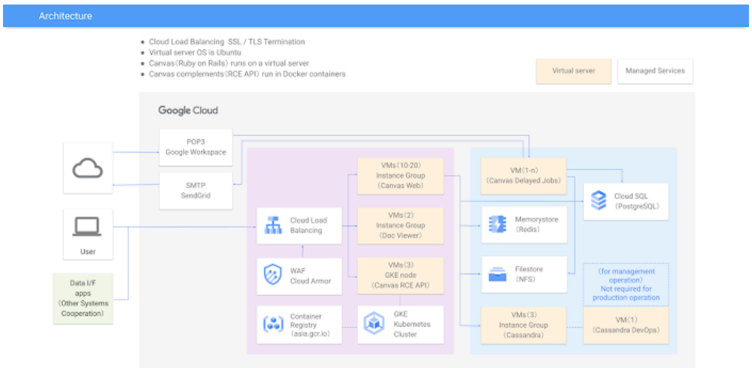
"We also chose Google Cloud because of the cost advantage and the solid technical support system. The latter was particularly vital to introducing the system within just three months. I have the impression that Google Cloud's support team is more than willing to work with us to solve our issues," shares Imahori.

Haruhiko Mitsuya, President of Bow Netsystems, and Yuki Ishikawa, System Engineer, who have worked on many Canvas LMS installations in Japan, had the following to say about building Canvas LMS on a Google Compute Engine VM:

"While Canvas LMS is designed for a specific cloud platform, it has the flexibility to run on-premises or on other cloud platforms without issue. However, we had never actually run Canvas LMS on Google Cloud, nor had we ever really used Google Cloud before. This was our first experience with it, and we were very impressed with its ease of use and higher-than-expected performance. The documentation was thorough, so we didn't expect any issues. Since the university has chosen Google Cloud as its platform, there is no reason to oppose the migration because it would be easier to manage in the long run," said Mitsuya.

"For this project, we took advantage of the cloud-native design of Canvas LMS to incorporate a number of managed services, which was one of the major advantages when running the system on Google Cloud," said Ishikawa. "Not only did we adopt Cloud Load Balancing (https://cloud.google.com/load-balancing) to achieve easy, automatic scaling configurations for web applications and Cloud SQL for PostgreSQL (https://cloud.google.com/sql/docs/postgres) as the core database for managing LMS data, we also adopted Memorystore (https://cloud.google.com/memorystore) for data caching, Filestore (https://cloud.google.com/filestore) for managing files such as teaching materials and assignments and Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine) for integrated containerized management of microservice applications."




*Enlarge Image*
(https://services.google.com/fh/files/misc/keio-univ-architecture-en.png)

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## Successfully launching an LMS in three months with Google Cloud's support team

The Canvas LMS environment was built on Google Cloud in anticipation of a September 2020 deadline. It was also the first time that Bow Netsystems, which has had extensive experience in implementing Canvas LMS, was taking on such a challenge. Despite this, Mitsuya says they were able to figure out how to use Google Cloud in a short period of time. He refers to their successful launch in September, with nearly 40,000 faculty, staff, and students now benefiting from state-of-the-art e-learning offered by this LMS.

"Without even the slightest delay permitted, we asked for support not only from Google Cloud representatives in Japan but also from the US. They were quick to respond, which really helped us out. If we had had to wait a week, we would not have been able to make this migration. I'm very grateful for their quick responses," shares Imahori.

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such as teaching materials and assignments and Google Kubernetes Engine for integrated containerized management of microservice applications."

*—Yuki Ishikawa, System Engineer, Bow Netsystems*

The system is currently still being operated in parallel with the old LMS. "We have confirmed that about half of the students have logged on in the six months since September," says Imahori. The university is already realizing many advantages of the Google Compute Engine system compared to the old on-premises LMS, such as its need for very little maintenance and operational stability even during peak periods of heavy access, such as during exam season. He adds that the teaching staff are gradually coming around to the system's convenience.

"Besides the fact that use has spread beyond the teaching staff who first used the new system, we held about 50 training sessions over two weeks during the early stages of implementation, and created training videos in cooperation with Bow Netsystems' Ishikawa. Many of our teaching staff are now using Canvas LMS," said Imahori. "It also seems that its use is spreading to universities other than Keio University through adjunct professors. The old LMS is expected to be retired in about a year."

"With this environment put in place, there has been an increasing need for some faculties to tightly manage study histories right up to state examinations. In fact, we at the ITC head office haven't been directly involved in education in any way until now, but I have a feeling that through this project, both we and the teaching staff will enter a new, different stage of our careers. I hope Google Cloud will continue offering advanced services to help this goal," concludes Takaharu Takeuchi, Director, Information Technology Center (ITC) Head Office, Keio University.

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